

Math Department, SUSTech
MA325, Numerical PDEs
Spring 2026
Syllabus

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1 General Information

1.1 Class time/location

Tue. 14:00 - 15:50 each odd week, 206 Business School Building,
Thu. 14:00 - 15:50 each week, 206 Business School Building.

1.2 Office Hours

M710, College of Science Bldg, Wed., 09:30AM-11:30AM; or make an appointment for other time.

1.3 Textbook & Reference

Finite Difference Methods for Ordinary and Partial Differential Equations: Steady-State and Time-Dependent Problems, by R. J. LeVeque. (Textbook)

Numerical Solution of Partial Differential Equations: An Introduction, by K. W. Morton, D. F. Mayers (Reference)

Numerical Solution of Differential Equations Introduction to Finite Difference and Finite Element Methods, by Z.L. Li, Z.H. Qiao and T. Tang) (Reference)

1.4 Calculators

Calculators are not allowed for exams or quizzes unless permitted.

2 Course Structure

2.1 Assignments: HW 28% (4% for each)

Homework will be collected in class. The homework due in any given week is the homework corresponding to the material of the previous week. **No late assignments will be accepted.** Best 7 (out of 8) scores are counted for credit.

Homework must be readable and should be **stapled**. Illegible scribbles will receive no credit from the grader. You are encouraged to attempt all the questions and discuss with your classmates. However, the write-up must be of your own.

2.2 Quizzes 18% (6% for each)

Basically, you will have a short in-class quiz every three weeks, 4 times in total. It usually takes 30 minutes, which only requires you to understand concepts and basic problem solving skills. **No makeups will be made for quizzes.** Only best 3 scores are counted for credit, which means you can miss one quiz or do poorly on one quiz.

2.3 Midterm 20%

One midterm is planned for Thursday, Apr. 21, tentatively. Makeups may be given in extraordinary cases, but only with documented reasons.

2.4 Final Exam 36%

The Final Exam will be given during the Final Exam Week.
The location and date of the final will be announced in class.

3 Tentative Course Outline

Chapter 1 Finite Difference Approximations

- 1.1 Truncation errors
- 1.2 Deriving finite difference approximations
- 1.3 Second order derivatives
- 1.4 Higher order derivatives
- 1.5 A general approach to deriving the coefficients

Chapter 2 Steady States and Boundary Value Problems

- 2.1 The heat equation
- 2.2 Boundary conditions
- 2.3 The steady-state problem
- 2.4 A simple finite difference method
- 2.5 Local truncation error
- 2.6 Global error
- 2.7 Stability
- 2.8 Consistency
- 2.9 Convergence
- 2.10 Stability in the 2-norm
- 2.11 Greens functions and max-norm stability
- 2.12 Neumann boundary conditions
- 2.13 Existence and uniqueness
- 2.14 Ordering the unknowns and equations
- 2.15 A general linear second order equation
- 2.18* Nonuniform grids, Adaptive mesh selection

Chapter 3 Elliptic Equations

- 3.1 Steady-state heat conduction

- 3.2 The 5-point stencil for the Laplacian
- 3.3 Ordering the unknowns and equations
- 3.4 Accuracy and stability
- 3.6 Other elliptic equations
- 3.7 Solving the linear system
- 3.8 Galerkin methods: spectral methods and finite element methods

Chapter 4 Solve the linear system

- 4.1 Fast solver for the linear system
- 4.2 Iterative methods
- 4.3 Convergence analysis
- 4.4 Applications for the linear system obtained from elliptic PDEs

Chapter 5–8 Topics on numerical solutions of ordinary differential equations

Brief review on ODE and numerical methods

Chapter 9 Diffusion Equations and Parabolic Problems

- 9.1 Local truncation errors and order of accuracy
- 9.2 Method of lines discretizations
- 9.3 Stability theory
- 9.4 Stiffness of the heat
- 9.5 Convergence, PDE versus ODE stability theory
- 9.6 VonNeumann analysis
- 9.7 Multidimensional problems
- 9.8 The locally one-dimensional method
- 9.8.1 Boundary conditions
- 9.8.2 The alternating direction implicit method
- 9.9* Other discretizations

Added topics Nonlinear parabolic problems: Allen-Cahn equations

Chapter 10 Advection Equations and Hyperbolic Systems

- 10.1 Advection
- 10.2 Method of lines discretization
- 10.2.1 Forward Euler time discretization
- 10.2.2 Leapfrog
- 10.2.3 Lax–Friedrichs
- 10.3 The Lax–Wendroff method
- 10.4 Upwind methods and The BeamWarming method
- 10.5 Stability analysis: Von Neumann analysis
- 10.6 Characteristic tracing and interpolation
- 10.7 The Courant–Friedrichs–Lewy condition

Added topics Nonlinear hyperbolic problems: conservation laws

Added topics* Deep Learning methods & Numerical PDE.

***Chapter: Selected topics**

***: covered in class, not required in any tests.**